From the Biquaternionic Carrier to Schrödinger at $\alpha \to 1^-$

Sam Leizerman • LIOR Research Initiative • ORCID 0009-0000-0133-2291

One-sheet companion to *Schrödinger Evolution from the Bare Causal Accumulation Law at $\alpha \to 1^-$*

Origin. CAL is not a theory of quantum gravity. It is an information-physics engine for non-Markovian reality, built to lift self-attention to a field-theoretic form from first principles of cognition, causality and intelligence. Its derivations of known physics are unit tests: a law of causal memory that cannot recover established physics has no business modeling minds or societies.

Definition 1 (Information). Information := distinguishability of physical states.

Remark 1 (Difference and observation). To distinguish physical states is to observe a difference between them. Change is a difference between states along an ordered evolution; thus change is temporally ordered distinguishability. Observation here denotes physical distinguishability, not a required conscious observer.

Definition 2 (Causality). Causality := change in physical state, including its rate of change.

From On Information Physics: From State Distinguishability to Causal Accumulation, Sec. 1.

The Causal Accumulation Law (On Information Physics, Eq. 61)

$$T^{\mu\nu a}{}_{\rho\sigma}(x) = \underbrace{\alpha_{\text{frac}}\, J^{\mu\nu a}{}_{\rho\sigma}(x)}_{\text{present source}} - (1-\alpha_{\text{frac}}) \int_{J^-(x)} \underbrace{k(\tau; x, x')}_{\text{history kernel}}\ \underbrace{\Pi^{\mu\nu}{}_{\rho\sigma|\alpha\beta}{}^{\gamma\delta}(x, x')}_{\text{slot coupling}}$$
$$\times \underbrace{P^{\alpha}{}_{\alpha'}(x,x')\, P^{\beta}{}_{\beta'}(x,x')\, P_{\gamma}{}^{\gamma'}(x,x')\, P_{\delta}{}^{\delta'}(x,x')}_{\text{spacetime transport of the past to } x} \times \underbrace{\Phi^{a}{}_{b}(x)\, P^{b}{}_{c'}(x,x')}_{\text{internal channel action}}\ J^{\alpha'\beta'c'}{}_{\gamma'\delta'}(x')\, d^4x'.$$

Read left to right: what the field is now (T) equals a share α_{frac} of the present source, minus a share $1-\alpha_{\text{frac}}$ of the past. Only the causal past $J^-(x)$ counts. Each past contribution is weighted by how long ago it happened (k), carried to the present point so it can be compared there (the four P factors), and rotated in its internal channel (ΦP). The single dial α_{frac} sets how much the present weighs against memory. Below, $\alpha \equiv \alpha_{\text{frac}}$ and the law is reduced to the spinless, gauge-neutral sector, where the transports are the identity.

Definitions and symbols

Sym.	Meaning	Type	Step
$T^{\mu\nu a}{}_{\rho\sigma}$	Accumulated output tensor	tensor field	CAL
$J^{\mu\nu a}{}_{\rho\sigma}$	Indexed source current	tensor field	CAL
$J^-(x)$, τ	Causal past of x; proper time	set; real	CAL
J	Carrier $\sum_a Z_a e_a$	$M_2(\mathbb{C})$	0
Z_a	Channel amplitude	complex	0
e_a	Basis $e_0 = I_2$, $e_j = -i\sigma_j$	$M_2(\mathbb{C})$	0
α_{frac}, ε	Interaction depth; $\varepsilon = 1-\alpha$	real	CAL
σ	Orientation of the source's past	± 1	4
k	History kernel	real	CAL
n, t	Update count; rest radians $t = \varphi_0 n$	integer; real	1
g, $\hat{g}$	Profile; its transform	real fn.	1

Sym.	Meaning	Type	Step
Π	Output/input slot coupling	tensor	CAL
$P^{\alpha}{}_{\alpha'}$	Parallel propagator	bitensor	CAL
$\Phi^{a}{}_{b}$	Action on internal index	matrix field	CAL
w_j	Memory weight at lag j	real	1
W^{α}, $W(z)$	Weyl integral; memory sum	op.; fn.	2, 3
θ	Channel angle $\alpha\pi/2$	angle	2
$F(z)$	Characteristic function of the loop	fn.	3
φ	Rotation $+\, i$ decay per update	complex	4
$\langle r^2 \rangle$	Second moment of g	real > 0	6
m	Mass, $\hbar/2m = \langle r^2 \rangle/12$	real > 0	6

0. The carrier

Each channel carries a Boltzmann-type decay weight and a Feynman-type phase weight; four channels assemble into one matrix.

$$Z_a = A_a e^{-H_a/\tau_E} + B_a e^{iS_a/\hbar_{\text{eff}}}, \qquad J = \textstyle\sum_{a=0}^{3} Z_a e_a \in M_2(\mathbb{C}).$$

1. The law as a feedback loop on the carrier

T is the new state; feed it back as the next current. P is the full real transport; n counts updates.

$$J_{n+1} = T_n = \alpha J_n - (1-\alpha) \int_{J^-} k\, P\, J.$$

On the spinless, gauge-neutral sector with a power-law kernel this reads

$$J_{n+1} = \alpha J_n - \varepsilon\, \hat{g}(k) \sum_{j \geq 1} w_j J_{n-j}, \qquad w_j = \frac{j^{\alpha} - (j-1)^{\alpha}}{\Gamma(\alpha+1)}.$$

J must be complex: the phase channel $B_a e^{iS_a/\hbar_{\text{eff}}}$ rules out a real carrier.

2. The memory acting on a wave

The past-only integral of an oscillation returns the same oscillation, rescaled and rotated.

$$W^{\alpha} e^{-iEt/\hbar} = (-iE/\hbar)^{-\alpha} e^{-iEt/\hbar}, \qquad 0 < \alpha < 1.$$

It converges only below $\alpha = 1$, which is why the limit is one-sided. The turn $\theta = \alpha\pi/2$ belongs to the source; nothing multiplies the memory.

3. The modes of the loop

A mode z^n solves the loop when $F(z) = 0$.

$$F(z) = z - \alpha + \varepsilon\, \hat{g}\, W(z), \qquad W(z) = \textstyle\sum_{j \geq 1} w_j z^{-j}.$$

With flat weights ($\alpha \to 1$) at $k = 0$: $z^2 - (1+\alpha)z + 1 = 0$, both roots exactly on $|z| = 1$.

4. The past pulls back: rotation

Below $\alpha = 1$ the memory enters with the sign $-(1-\alpha) < 0$, so the accumulated past pulls the current back, and the current rotates:

$$\varphi = \sqrt{\varepsilon \hat{g}} + i\Big[\tfrac{1}{2}\varepsilon(1-\hat{g}) + \tfrac{\pi}{4}\varepsilon^{3/2}\sqrt{\hat{g}}\Big] + \dots$$

Its direction (σ) is inherited from the source's past.

5. Unitarity is derived, in the limit

For $\alpha < 1$ each mode is a contraction:

$$\exp\Big[-\big(i + \sqrt{1-\alpha}\big)\tfrac{\hbar}{2m}k^2 t - \tfrac{\pi}{4}(1-\alpha)\, t\Big].$$

Both vanish as $\alpha \to 1^-$, where the evolution is a strongly continuous unitary group.

6. Take the limit

Let $\varepsilon \to 0^+$, with t in radians of the carrier's own rotation. The rotation slows with wavenumber as $\sqrt{\hat{g}}$:

$$\big(1 - \sqrt{\hat{g}(k)}\big)\, t = \frac{\langle r^2 \rangle}{12} k^2 t + O(k^4), \qquad \frac{\hbar}{2m} = \frac{\langle r^2 \rangle}{12}.$$

The co-rotating branch has positive mass, set by the kernel's spatial width.

7. The kinetic term, exactly

Take wavelengths long against the kernel's spatial width at fixed $\hbar/2m$. Higher orders vanish and the rest rotation is only a global phase.

$$\boxed{i\hbar\, \partial_t J = -\frac{\hbar^2}{2m} \nabla^2 J}$$

8. Why it is Markov

As $\alpha \to 1^-$ the accumulated past of a rotating current lags it by a quarter turn, $W \approx 1/(i\varphi)$. Pulling back along that lag turns the current instead of shrinking it. That quadrature is the i in $i\hbar\partial_t$.

As $w_j \to 1$ the kernel becomes a flat accumulator whose inverse is ∂_t, and the hidden memory modes collapse onto the carrier itself: Markov through the fundamental theorem of calculus, no records, no enlarged state space. ***[Derived]***

9. The readout survives

On a one-way past the Born normalization is conserved at the edge; with both directions present it swings.

$$\tfrac{1}{2}\,\mathrm{tr}(JJ^{\dagger}) = \textstyle\sum_a |Z_a|^2 \quad \text{is constant in } t.$$

10. Which sector carries the dynamics

Linearized, $Z_a(t) = A_a e^{\lambda_a t} + B_a e^{i\omega_a t}$ with $\lambda_a > 0$. Boundedness on the whole time line removes the decay sector, so the phase sector $B_a e^{iS_a/\hbar_{\text{eff}}}$ is what becomes Schrödinger. Below the edge it survives as the $\sqrt{1-\alpha}$ admixture.

11. What the edge is

Galilean (time ordering only; the Lorentz signature belongs to the $\alpha = 2$ closure), free, and single-carrier: it can carry existing entanglement but cannot create it. Above $\alpha = 1$ the one-way equation has no causal, bounded continuation.